Duplicate Contamination, Schema Loss, and Cross-Cohort Generalization in Public Cardiovascular Risk Benchmarks: A Reproducibility Audit and a Schema-Flexible Tabular Transformer

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***Abstract*—Public cardiovascular-risk benchmarks are routinely used to report accuracies above 90%, yet duplicated records, cross-cohort patient overlap, and schema loss during dataset merging receive little scrutiny. We audit six open data sources (74,013 non-overlapping patients after quality control) and show that (i) 723 of 1,025 rows (70.5%) in a widely used Kaggle heart-disease copy are exact duplicates, so a random 80/20 split places 97.1% of test rows in the training set and inflates accuracy to 98.5%, versus 77.1% after deduplication; (ii) all 270 UCI Statlog records duplicate Cleveland records; (iii) inner-join harmonization discards unshared features and drives zero-shot cross-cohort AUROC down to 0.535–0.713; and (iv) duplicates are too few (1.02%) to explain a published 90.1% accuracy on a 71,025-row pool, where a 70/15/15 reproduction gives about 73% with or without them. We propose SchemaFlexTabNet-QC, a feature-tokenizing transformer that models only the variables each cohort actually observes, trained with feature dropout, an auxiliary hypertension concept head and a monotonicity penalty, and deployed with temperature-scaled conformal abstention. Under leave-one-cohort-out (LOCO) evaluation it reaches a mean AUROC of 0.763, versus 0.611 for inner-join LightGBM and 0.740 for NaN-aware union-schema LightGBM; against the latter, no per-cohort difference is significant after Holm correction. At 50% test-time missingness it keeps 0.712 AUROC versus 0.613 for native-NaN LightGBM. Conformal sets calibrated on source cohorts under-cover on held-out hospitals (84.2–88.7% against a nominal 90%). Outcome definitions differ across cohorts, and seed-to-seed variance is not yet quantified.**

***Index Terms*—Cardiovascular risk prediction, tabular transformers, data leakage, multi-cohort harmonization, missing data, conformal prediction, reproducibility.**

I. Introduction

CARDIOVASCULAR disease (CVD) causes roughly 17.9 million deaths per year worldwide. Machine-learning risk models are attractive for screening, but accuracies published on public benchmarks frequently exceed what noisy routine records can support, and they rarely survive transfer to a new hospital. Two well-documented sources of optimism are leakage between training and test data [6] and near-duplicate records in benchmark files [7]. A third, specific to multi-source clinical work, is the harmonization step itself: merging cohorts through an inner join keeps only the columns present everywhere and silently discards the most diagnostic variables of specialist cohorts (chest-pain type, exercise ECG response, vessel counts).

A recent study, HybridTabNet-QC [1], reports 90.1% accuracy and 93.6% [verify AUROC: the project reports give 0.963] AUROC on a merger of two Kaggle datasets, using a transformer encoder fused with engineered meta-features (BMI, pulse pressure, and a cholesterol–glucose product). Motivated by the gap between that figure and what we observe on clean data, this paper makes four contributions:

1) **A data audit** of six public sources quantifying exact-duplicate contamination, cross-cohort patient overlap and physiologically implausible records, with a deterministic plausibility gatekeeper and a reproducible audit log.
2) **A reproducibility analysis** of [1] that documents internal inconsistencies, bounds what the duplicate issue can explain, and reports a 70/15/15 reproduction of the published setup.
3) **SchemaFlexTabNet-QC**, a schema-flexible tokenizing transformer that trains across cohorts with non-identical columns, with physiological regularizers and conformal abstention.
4) **A leave-one-cohort-out benchmark** against strong baselines, including a NaN-aware union-schema gradient-boosting model, with Holm-corrected DeLong tests, ablations, missingness stress tests against native-NaN trees, a conformal cohort-shift audit, and subgroup, federated and decision-curve analyses.

The paper is organized as follows. Section II reviews related work, Section III describes data and quality control, Section IV presents the audit, Section V the method, Section VI the protocol, Section VII the results, Section VIII the discussion and limitations, and Section X concludes.

II. Related Work

Tabular deep learning. FT-Transformer [8], TabNet [9] and SAINT [10] show that attention over feature tokens is competitive on tabular data, although gradient-boosted trees such as LightGBM [11] remain strong baselines. Our tokenizer follows the feature-token idea but treats the *set of observed features* as the input, so unobserved columns are masked rather than imputed.

Domain knowledge in models. Concept bottleneck models [12] supervise intermediate, human-interpretable concepts,

and monotonic models [13] encode the prior that risk should not fall as a harmful marker rises. We use both as soft regularizers rather than hard architectural constraints.

Calibration and abstention. Temperature scaling [14] and conformal prediction [15], [16] provide calibrated probabilities and set-valued predictions with finite-sample coverage under exchangeability. That assumption fails under cohort shift [17], [18], which matters for our setting and which we measure directly (Section VII-D).

Reporting standards. TRIPOD+AI [22] and leakage checklists [6] motivate the audit-first design of this work.

III. Data and Quality Control

A. Sources

We use six public sources: UCI Heart Disease from four institutions (Cleveland, Hungarian, Switzerland, VA Long Beach) [2], [3]; the Kaggle heart-disease copy (1,025 rows); the Sulianova cardiovascular dataset (70,000 examination records); a Framingham teaching subset (4,240 records) [5]; UCI Statlog Heart (270 records); and the Z-Alizadeh Sani angiography cohort (303 records; 56 clinical and laboratory variables) [4].

B. Canonical schema

A registry defines 22 union features: continuous variables (age, systolic and diastolic blood pressure, BMI, pulse pressure, cholesterol, maximum heart rate, ST depression), categorical variables (sex, ATP-III cholesterol tier, glucose tier, chest-pain type, exercise angina, smoking), and one auxiliary concept, Stage 2 hypertension (SBP $\geq$ 140 or DBP $\geq$ 90 mmHg). Variables that a cohort does not record are left *absent*, not zero-filled.

C. Plausibility gatekeeper

Instead of a statistical $|Z| > 3$ filter, records are rejected if they violate physiological bounds (age 18–110 y; SBP 60–260; DBP 40–160 mmHg; BMI 12–70; cholesterol 80–650 mg/dL; heart rate 50–230 bpm; ST depression 0–10 mm) or hemodynamic consistency (SBP $-$ DBP $\geq$ 10 mmHg). Exact patient-level duplicates are removed. Table II gives the audit log. The eight clean cohorts sum to 74,283 records; excluding the 270 Statlog rows that duplicate Cleveland leaves 74,013 non-overlapping patients, which is the pool used throughout the paper. Sulianova contributes 92.6% of these (68,568 of 74,013), a fact that shapes several conclusions below.

IV. Reproducibility Audit

A. Duplicate leakage and cohort overlap

The Kaggle heart copy contains only 302 unique patients; the remaining 723 rows (70.5%) are exact copies. Under a random 80/20 split, 199 of 205 test rows (97.1%) have an identical twin in the training set. Evaluating on that split yields near-perfect scores that vanish once duplicates are removed (Table III). A signature match on age, sex, resting blood pressure, cholesterol, peak heart rate and ST depression further shows that all 270 Statlog patients are identical to Cleveland patients, so training on one and testing on the other is leakage; we excluded the overlap from every cross-cohort experiment.

B. Observations on the published HybridTabNet-QC results

We did not have access to code or splits for [1], and we do not claim to know why its figures differ from ours. We report only what can be checked from the article itself, plus a reproduction of its stated split.

- **Inconsistent confusion matrices.** The text describes a matrix with 980/1,100 negatives and 990/1,100 positives correct (accuracy 89.5%), whereas Fig. 2 of that article shows 4,795/533/395/4,930 on 10,653 samples, implying accuracy 91.3% and positive-class recall 0.926, against 0.90 in its Table 1. [re-verify the Fig. 2 cell counts against the published figure; the project reports list a different matrix (5,044/4,688/444/477)]
- **Class balance.** The test set is described as balanced (5,328 vs. 5,325), while the preprocessing section describes a roughly 1:2 minority-to-majority imbalance with class weights 0.45/0.55.
- **Test-set size.** 10,653 equals 15% of 71,025, the sum of the raw rows of the two Kaggle files ($1{,}025 + 70{,}000$). This is consistent with the duplicated 1,025-row copy having been used, and with little row removal despite the described outlier filtering.
- **Limits of the duplicate explanation.** The 723 duplicate rows are $723/71{,}025 = 1.02\%$ of the pooled set. Even if every duplicate were classified correctly, raising their accuracy from about 73% to 100% would lift overall accuracy by only about $0.0102 \times (1 - 0.73) \approx 0.3$ percentage points. Duplicates therefore cannot by themselves explain 90.1% accuracy on a pool that is 98.6% Sulianova, where well-tuned models in our benchmark reach roughly 73–76%.
- **Engineered features are not architecture-specific.** Adding BMI, pulse pressure and tier features changes AUROC by $+0.0014$ (MLP) to $+0.0124$ (logistic regression) across seven model families (Section VII-A). Work [1] states that baselines were retrained under the same pipeline, so we do not allege unequal treatment; we show only that the features help every model.

C. Reproduction of the 70/15/15 setup

We applied the stated 70/15/15 train/validation/test protocol to (a) the raw 71,025-row pool, (b) the same pool with the 723 Kaggle duplicates removed, and (c) Sulianova alone, using the same hemodynamic filtering in each case and a standard supervised pipeline [name the model and settings used for this reproduction]. Table IV shows that all three configurations land at about 73% accuracy and 0.795–0.799 AUROC. Keeping the duplicates changes accuracy by $+0.30$ points, in line with the bound above. We could not reproduce 90.1% with these features and this protocol. The discrepancy may reflect preprocessing or reporting details that are not documented in the article; we cannot tell which, and we recommend release of splits and code.

V. Method: SchemaFlexTabNet-QC

Fig. 1 summarizes the architecture.

TABLE I
COHORT SUMMARY. "CARDIOLOGY VARIABLES" DENOTES CHEST-PAIN TYPE, PEAK HEART RATE, EXERCISE ANGINA, ST DEPRESSION AND VESSEL COUNT. OUTCOME DEFINITIONS DIFFER ACROSS SOURCES AND ARE MAPPED TO ONE BINARY LABEL.

Source	Clean N	Outcome construct	Cardiology variables	Role in this study
Sulianova CVD	68,568	Presence of CVD at examination	Not recorded	Training (all LOCO folds)
Framingham subset	4,238	Incident CHD during follow-up	Not recorded	Training; held-out unit
UCI Cleveland	303	Angiographic coronary disease	Recorded	Training; held-out unit
UCI Hungarian	293	Angiographic coronary disease	Recorded (partly missing)	Training; held-out unit
UCI VA + Switzerland	308	Angiographic coronary disease	Recorded (partly missing)	Training; held-out unit
Z-Alizadeh Sani	303	Coronary stenosis on angiography	Recorded (plus lab panel)	Training; held-out unit
UCI Statlog	270	Angiographic coronary disease	Recorded	Excluded (duplicates Cleveland)
Kaggle heart copy	302 unique	Angiographic coronary disease	Recorded	Audit only

TABLE II
QUALITY-CONTROL AUDIT LOG.

Source	Raw	Dup.	Implaus.	Clean
UCI Cleveland	303	0	0	303
UCI Hungarian	294	1	0	293
UCI Switzerland	123	0	11	112
UCI VA Long Beach	200	3	1	196
Sulianova CVD	70,000	58	1,374	68,568
Framingham subset	4,240	0	2	4,238
UCI Statlog	270	0	0	270
Z-Alizadeh Sani	303	0	0	303
Kaggle heart copy	*1,025*	*723*	*0*	*302*

TABLE III
EFFECT OF DUPLICATE LEAKAGE ON A SMALL CARDIOLOGY BENCHMARK.

Setup	Data	Acc.	AUROC
Random split, duplicates kept	Kaggle copy (1,025)	98.54%	0.9980
Deduplicated patient split	Cleveland (302)	77.05%	0.8593
Examination data only	Sulianova (68,568)	73.21%	0.7943

TABLE IV
REPRODUCTION OF THE PUBLISHED 70/15/15 SETUP. HEMODYNAMIC FILTERING REDUCES THE ROW COUNTS.

Configuration	Clean N	Test N	Acc. (%)	AUROC
Pool, duplicates kept (71,025 raw)	69,692	10,454	73.48	0.7978
Pool, deduplicated (70,302 raw)	68,969	10,346	73.18	0.7954
Sulianova alone (70,000 raw)	68,667	10,301	73.42	0.7987

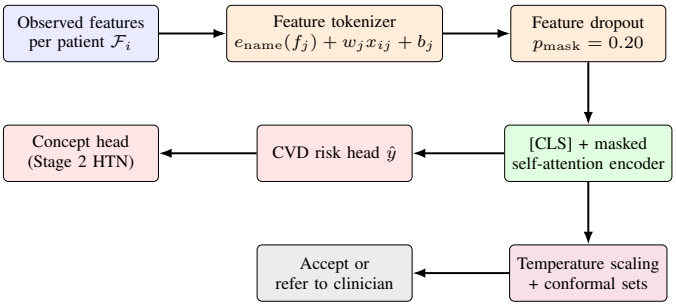


Fig. 1. SchemaFlexTabNet-QC pipeline. Only observed features are tokenized; absent features are masked in attention. A monotonicity penalty acts on $\partial \hat{y}/\partial x_k$ during training.

A. Schema-flexible tokenizer

For patient i with observed feature set $\mathcal{F}_i$, each observed feature $j \in \mathcal{F}_i$ becomes a token

$$z_{ij} = e_{\text{name}}(f_j) + w_j\, x_{ij} + b_j, \tag{1}$$

where $e_{\text{name}}(f_j) \in \mathbb{R}^d$ is a learned feature-identity embedding and (w_j, b_j) is a feature-specific linear projection (categorical variables use learned value embeddings). Continuous inputs are z-scored with training statistics. A learnable `[CLS]` token is prepended and a key-padding mask removes unobserved features from attention, so no inner join, zero-fill or imputation is needed. The encoder has 2 layers, 4 heads, width $d = 64$, feed-forward width 128 and dropout 0.10.

B. Feature dropout

During training each observed token is dropped with probability $p_{\text{mask}} = 0.20$, simulating hospitals that supply only part of the panel and forcing the encoder not to rely on any single variable.

C. Concept head and monotonicity

The `[CLS]` representation also predicts Stage 2 hypertension, giving $\mathcal{L}_{\text{concept}}$ (binary cross-entropy). Monotonicity is encouraged in two ways for the set of harmful markers $\mathcal{M} = \{\text{age, SBP, cholesterol, ST depression}\}$ [confirm set against code; an earlier draft also listed DBP and pulse pressure]. First, the logit contains a residual branch with softplus-constrained, hence non-negative, weights:

$$\hat{z}_i = \mathbf{w}_{\text{out}}^{\top}\mathbf{h}_i^{\text{CLS}} + b_0 + \sum_{k\in\mathcal{M}} \text{softplus}(w_k)\, x_{ik}\, \mathbb{I}[x_{ik} \text{ observed}], \tag{2}$$

with $\hat{y}_i = \sigma(\hat{z}_i)$. Second, an input-gradient penalty computed by automatic differentiation,

$$\mathcal{L}_{\text{mono}} = \frac{1}{|\mathcal{M}|} \sum_{k\in\mathcal{M}} \text{ReLU}\left(-\frac{\partial \hat{y}_i}{\partial x_{i,k}}\right) \mathbb{I}[x_{i,k} \text{ observed}]. \tag{3}$$

The total loss is

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \lambda_1 \mathcal{L}_{\text{concept}} + \lambda_2 \mathcal{L}_{\text{mono}}, \tag{4}$$

with $\lambda_1 = 0.10$ and $\lambda_2 = 1.20$. Because the attention path is unconstrained, neither mechanism guarantees monotonicity; we therefore *measure* violations (Section VII-C).

D. Calibration and abstention

Temperature scaling is fitted by L-BFGS on held-out validation data. Split conformal prediction at $\alpha = 0.10$ then yields a label set $\mathcal{C}(x) \subseteq \{0, 1\}$. If $|\mathcal{C}(x)| = 1$ the prediction is accepted; otherwise the case is flagged for clinician review. A Mondrian variant computes non-conformity quantiles separately per sex.

E. Conformal procedure

Let $p(x)$ be the temperature-scaled probability of disease and define the non-conformity score $s(x, y) = 1 - p_y(x)$. On a calibration set of n labeled patients, let $\hat{q}$ be the $\lceil (n + 1)(1 - \alpha) \rceil / n$ empirical quantile of the scores $s(x_i, y_i)$. The prediction set is

$$\mathcal{C}(x) = \{y \in \{0, 1\} : \; s(x, y) \leq \hat{q}\}. \tag{5}$$

Under exchangeability of calibration and test data, $\Pr[Y \in \mathcal{C}(X)] \geq 1 - \alpha$. The Mondrian variant computes $\hat{q}_g$ separately for each sex g, which targets the group-conditional guarantee $\Pr[Y \in \mathcal{C}(X) \mid G = g] \geq 1 - \alpha$, again under within-group exchangeability. A case is *accepted* when $|\mathcal{C}(x)| = 1$, and *referred* when $\mathcal{C}(x) = \{0, 1\}$ or $\emptyset$.

We keep two quantities strictly apart. *Set coverage* is $\Pr[Y \in \mathcal{C}(X)]$, the quantity the conformal guarantee is about. The *acceptance rate* is the fraction of patients with $|\mathcal{C}(x)| = 1$; it is a selective-prediction rate and carries no coverage guarantee. In leave-one-cohort-out use, calibration scores come from source cohorts while test patients come from a different hospital, so $P_{\text{source}}(X, Y) \neq P_{\text{target}}(X, Y)$ and exchangeability does not hold; we measure the resulting coverage loss in Section VII-D.

F. Schema-aware federated averaging

In the federated simulation each hospital node h observes a feature subset $\mathcal{F}_h$. Parameters shared across features (transformer blocks, `[CLS]`, heads) are averaged over all nodes, weighted by local sample size n_h:

$$\theta_{\text{shared}} \leftarrow \sum_h \frac{n_h}{\sum_{h'} n_{h'}} \theta^{(h)}_{\text{shared}}. \tag{6}$$

Feature-specific parameters $\theta_j = (e_{\text{name}}(f_j), w_j, b_j)$ are averaged only over the nodes $H_j = \{h : j \in \mathcal{F}_h\}$ that observe feature j:

$$\theta_j \leftarrow \sum_{h \in H_j} \frac{n_h}{\sum_{h' \in H_j} n_{h'}} \theta^{(h)}_j. \tag{7}$$

This prevents embeddings of a feature from being diluted by nodes that never see it.

G. Evaluation metrics

Calibration is summarized by the expected calibration error over B equal-width probability bins,

$$\text{ECE} = \sum_{b=1}^{B} \frac{|S_b|}{N} \left| \text{acc}(S_b) - \text{conf}(S_b) \right|, \tag{8}$$

and by the Brier score $\frac{1}{N} \sum_i (\hat{y}_i - y_i)^2$. Clinical utility at decision threshold p_t is the net benefit

$$\text{NB}(p_t) = \frac{TP}{N} - \frac{FP}{N} \cdot \frac{p_t}{1 - p_t}, \tag{9}$$

compared with the treat-all strategy $\text{NB}_{\text{all}} = \pi - (1 - \pi)\, p_t / (1 - p_t)$, where π is prevalence [20].

VI. Experimental Protocol

Evaluation partitions. Each experiment uses one stated partition. (i) The pooled benchmark is stratified five-fold cross-validation on the 74,013 non-overlapping patients. (ii) The LOCO and DeLong analyses hold out each of five units in turn: UCI Hungarian (N=293), UCI VA+Switzerland (308), UCI Cleveland deduplicated (303), Z-Alizadeh Sani (303) and Framingham (4,238); the model trains on the remaining non-overlapping cohorts, and Statlog is excluded throughout. (iii) The missingness stress test and the ablation use a held-out 15% stratified multi-cohort test partition, so their values are not comparable with the LOCO table. (iv) Subgroup, decision-curve and bootstrap analyses use the 270-patient Statlog/Cleveland-overlap evaluation (model trained without Cleveland); these patients are a subset of Cleveland, so these results are not independent of the Cleveland LOCO row.

Baselines. (a) Inner-join LightGBM on shared features; (b) union-schema LightGBM with native missing-value handling; (c) HybridTabNet-QC *re-implemented from the published description* with an inner-join merge, since no official code is available. For missingness stress tests we add native-NaN LightGBM and XGBoost and median-imputed LightGBM. A pooled five-fold benchmark covers seven classical and neural models with and without engineered meta-features.

Metrics and tests. AUROC, accuracy, Brier score and expected calibration error (ECE). Paired AUROC differences use DeLong's test [19] with Holm–Bonferroni correction across the five held-out cohorts; clinical utility is assessed with decision-curve analysis [20]; federated experiments follow FedAvg [21].

Remaining statistical work. Table values are single runs. Bootstrap intervals (1,000 patient resamples) are reported only for the 270-patient analysis, and mean ± standard deviation over at least five seeds for the neural models remains outstanding.

Implementation. The encoder configuration is given in Section V. Optimizer, learning rate, epochs and seeds: [insert from configuration files]. Code and configuration are in the repository listed under Data and Code Availability.

TABLE V
POOLED 5-FOLD BENCHMARK: AUROC WITH RAW FEATURES (A) AND WITH ENGINEERED META-FEATURES (B); ACCURACY FOR B.

Model	A	B	Δ	Acc. B (%)
Logistic regression	0.7554	0.7678	+0.0124	70.19
Random forest	0.8195	0.8244	+0.0049	75.85
XGBoost	0.8186	0.8247	+0.0061	75.95
LightGBM	0.8187	0.8249	+0.0062	75.98
CatBoost	0.8152	0.8215	+0.0063	75.60
MLP	0.8096	0.8110	+0.0014	74.86
TabNet	0.8078	0.8105	+0.0027	74.87

VII. RESULTS

A. Pooled benchmark

On pooled stratified five-fold cross-validation (74,013 patients), LightGBM with raw plus engineered features reaches AUROC 0.8249 and accuracy 75.98% (75.39% with raw features only). XGBoost, random forest and CatBoost lie within 0.004 AUROC, MLP and TabNet reach about 0.811, and logistic regression 0.768 (Table V). Among the models tested these values form an empirical ceiling for this pool, and they are far below 90%. Engineered features help every model by between +0.0014 and +0.0124 AUROC.

B. Zero-shot cross-cohort transfer

Table VI and Fig. 2 show LOCO transfer. Inner-join models are weak (mean AUROC 0.611 for LightGBM and 0.645 for the re-implemented HybridTabNet-QC) because the shared columns exclude chest-pain type, exercise ECG variables and vessel counts. Retaining the union schema is the decisive change: NaN-aware LightGBM already reaches 0.740. SchemaFlexTabNet-QC reaches 0.763, a mean gain of +0.023 over that stronger baseline, driven by Z-Alizadeh (+0.102) and Hungarian (+0.022); it is lower on VA+Switzerland (−0.004) and Cleveland (−0.013).

Table VII gives paired DeLong tests. Against the re-implemented inner-join HybridTabNet-QC, SchemaFlexTabNet-QC is significantly better on Hungarian ($p < 10^{-10}$), VA+Switzerland (p=5.6×10^{-4}) and Cleveland (p=2.1×10^{-6}), and not significantly different on Z-Alizadeh (p=0.84) or Framingham (p=0.71). Against the union-schema LightGBM, the most informative comparison, the Z-Alizadeh gain is nominally significant (raw p=0.037) but no cohort survives Holm correction (adjusted $p \geq 0.18$). We therefore treat SchemaFlexTabNet-QC as statistically comparable to union-schema LightGBM on these cohorts, not superior to it.

C. Ablation and monotonicity

Table VIII isolates each component on the held-out 15% multi-cohort partition. The tokenizer produces by far the largest effect (+0.222 AUROC from variant 1 to 2). Feature dropout adds +0.004 AUROC and lowers monotonicity violations from 19.2% to 4.2%. The concept head changes AUROC by only +0.0007 and raises violations to 13.0%, so we do not claim a discrimination benefit from it; any value it has is interpretive rather than predictive. Adding the monotone skip path, the gradient penalty ($\lambda_2 = 1.20$) and temperature scaling gives the best accuracy (80.20%), ECE (0.057) and a violation rate of 0.26% on synthetic +0.5 SD perturbations of SBP and cholesterol. These are single-run figures without confidence intervals, and the evaluation split differs from that of Table VI.

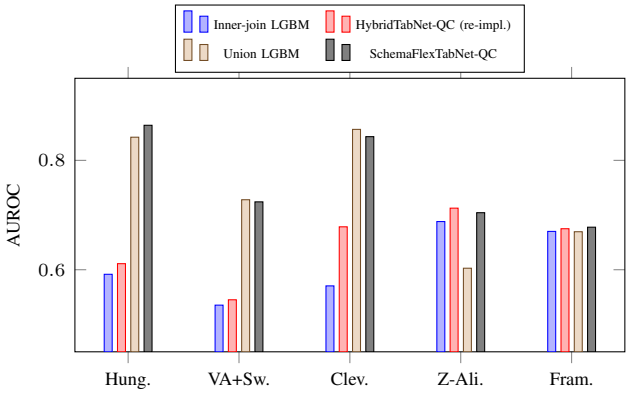


Fig. 2. Zero-shot AUROC by held-out cohort (data of Table VI).

D. Conformal abstention and cohort shift

At $\alpha = 0.10$ the model accepts 36.6–78.9% of patients depending on the cohort (Table IX). On accepted cases AUROC rises by 0.03–0.09 and accuracy by 5–10 points on four cohorts; on Framingham accuracy does not improve (51.3% to 51.2%). The accepted fraction is a *selective-prediction rate*, not conformal coverage.

Set coverage tells a different story. With calibration scores taken from source cohorts, empirical coverage $\Pr[Y \in \mathcal{C}(X)]$ on the held-out hospital falls to 84.2–88.7%, below the nominal 90%, confirming that exchangeability is violated. Recalibrating on a labeled slice of the target cohort restores coverage to 90.0–90.4%. This repair needs labeled target data, so it is not available in a true zero-shot deployment. On the Statlog/Cleveland-overlap analysis, the only one with bootstrap intervals, AUROC is 0.882 (95% CI 0.839–0.918) at full coverage; this is a 270-patient subset of Cleveland trained without Cleveland and is not independent of the 0.843 Cleveland LOCO figure.

E. Robustness to missing data

Table X and Fig. 3 show AUROC as 0–50% of features are masked at test time, on the held-out 15% partition. SchemaFlexTabNet-QC degrades from 0.852 to 0.712 and an unregularized tabular transformer from 0.838 to 0.699. The tree baselines now use native missing-value handling: at 50% missingness SchemaFlexTabNet-QC exceeds native-NaN LightGBM by +0.099 AUROC, native-NaN XGBoost by +0.078 and median-imputed LightGBM by +0.079. These margins are smaller than those against median-imputed trees reported in earlier drafts, where the gap reached 0.15. Two cautions apply. At 0% missingness the tree baselines (about 0.73) sit well below the pooled cross-validated LightGBM of Table V (0.825), so the partitions and tuning differ, and

TABLE VI
ZERO-SHOT LOCO AUROC. HYBRID = RE-IMPLEMENTED HYBRIDTABNET-QC WITH INNER JOIN; UNION LGBM = NAN-AWARE LIGHTGBM ON THE UNION SCHEMA. THE MEAN ROW IS THE UNWEIGHTED MEAN OF THE FIVE PER-COHORT VALUES. BOLD MARKS THE BEST VALUE PER ROW.

Held-out cohort (N)	Inner-join LGBM	Hybrid (re-impl.)	Union LGBM	SchemaFlexTabNet-QC	Δ vs. union
Hungarian (293)	0.5917	0.6111	0.8423	**0.8641**	$+0.0218$
VA+Switzerland (308)	0.5353	0.5451	**0.7280**	0.7241	-0.0039
Cleveland dedup. (303)	0.5705	0.6785	**0.8566**	0.8433	-0.0133
Z-Alizadeh Sani (303)	0.6881	**0.7126**	0.6028	0.7043	$+0.1015$
Framingham (4,238)	0.6701	0.6750	0.6695	**0.6778**	$+0.0083$
Unweighted mean	0.6111	0.6445	0.7398	0.7627	$+0.0229$

TABLE VII
PAIRED DELONG TESTS FOR SCHEMAFLEXTABNET-QC. "UNION" = UNION-SCHEMA NAN-AWARE LIGHTGBM, WITH HOLM–BONFERRONI ADJUSTMENT ACROSS THE FIVE COHORTS. "HYBRID" = RE-IMPLEMENTED INNER-JOIN HYBRIDTABNET-QC (UNADJUSTED p). N.S. = NOT SIGNIFICANT AT 0.05.

Held-out cohort	N	Δ vs. union	DeLong z	Raw p	Holm p	Sig. (Holm)	p vs. Hybrid
Hungarian	293	$+0.0218$	1.413	0.158	0.631	n.s.	$< 10^{-10}$
VA+Switzerland	308	-0.0039	-0.255	0.798	1.000	n.s.	5.6×10^{-4}
Cleveland dedup.	303	-0.0133	-0.760	0.447	1.000	n.s.	2.1×10^{-6}
Z-Alizadeh Sani	303	$+0.1015$	2.087	0.037	0.185	n.s.	0.84 (n.s.)
Framingham	4,238	$+0.0083$	0.722	0.470	1.000	n.s.	0.71 (n.s.)

TABLE VIII
COMPONENT ABLATION ON THE HELD-OUT 15% MULTI-COHORT PARTITION. VIOL. = COUNTERFACTUAL MONOTONICITY VIOLATIONS (%).

Variant	AUROC	Acc.	ECE	Viol.
1. Inner-join, shared feats.	0.6342	59.73	0.1368	15.04
2. + Schema tokenizer	0.8560	75.77	0.1672	19.23
3. + Feature dropout	0.8604	77.13	0.1008	4.18
4. + Concept head	0.8611	77.82	0.1081	12.97
5. + Mono. + temperature	0.8652	80.20	0.0572	0.26

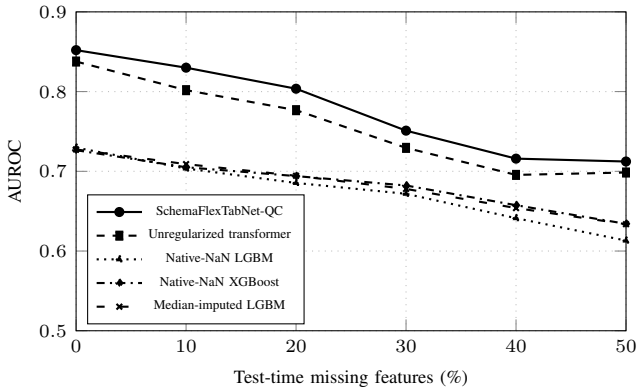


Fig. 3. Structured-missingness stress test (data of Table X).

the absolute gap should be read with care. And the 0.014–0.032 margin over the unregularized transformer is attributed to feature dropout but has not been separated from other differences. The unregularized-transformer series is from an earlier run than the tree series.

F. Fairness, federation and clinical utility

Subgroups. On the 270-patient overlap cohort, male and female false-positive rates are 0.398 and 0.119 and sensitivities 0.86 and 0.80; the age > 65 group has only 30 patients. In a separate 298-patient evaluation, marginal conformal calibration gave FPRs of 0.038 (male) and 0.186 (female); Mondrian calibration moved them to 0.173 and 0.246. It therefore *halved* the inter-sex FPR gap, from 0.149 to 0.072 (a 51% reduction), and lowered male FNR from 0.395 to 0.250, but it did not eliminate the gap and it raised the FPR of both groups. Subgroup samples are too small for confidence intervals to support firm conclusions.

Federated simulation. In a four-node simulation, schema-aware averaging (feature-specific parameters averaged only across nodes that observe the feature) reaches 0.767 AUROC after 10 rounds, versus 0.746 for isolated models and 0.875 for centralized training. Federation closes 16.3% of the gap between isolated and centralized training. It shares only model weights, and no formal privacy mechanism (differential privacy, secure aggregation) was applied, so "no row sharing" should not be read as a privacy guarantee.

Decision curves. On the same 270-patient cohort (prevalence 44.4%), net benefit exceeds treat-all at thresholds from 0.10 to 0.30, e.g. 0.346 vs. 0.306 at 0.20, equivalent to about four additional net true-positive decisions per 100 patients. This holds for one high-prevalence single-center cohort and should not be generalized.

VIII. DISCUSSION AND LIMITATIONS

What the evidence supports. Duplicate contamination and cross-cohort overlap are real and large in popular files, and they invalidate random-split results on them. Inner-join harmonization is costly for transfer across cohorts with different measurement panels. A union-schema representation recovers most of that loss, whether implemented as a masked transformer or as NaN-aware trees; the transformer adds a modest mean gain that is not statistically distinguishable from the tree baseline on any single cohort after multiplicity correction, a

TABLE IX
CONFORMAL ABSTENTION AT $\alpha = 0.10$ UNDER COHORT SHIFT. ACCEPTED = FRACTION WITH A SINGLETON SET. COVERAGE = $\Pr[Y \in \mathcal{C}(X)]$ WITH NOMINAL VALUE 90%, USING SOURCE-COHORT CALIBRATION (SRC.) OR A LABELED SLICE OF THE TARGET COHORT (TGT.). AUROC AND ACCURACY: ALL PATIENTS → ACCEPTED PATIENTS.

Held-out cohort	Accepted	Src. coverage	Tgt. coverage	AUROC	Accuracy (%)
Hungarian	78.2%	88.7%	90.4%	.864 → .893	76.8 → 85.2
VA+Switzerland	72.7%	86.4%	90.3%	.724 → .771	74.0 → 83.5
Cleveland dedup.	78.9%	88.1%	90.1%	.843 → .876	73.6 → 78.7
Z-Alizadeh Sani	36.6%	84.2%	90.1%	.704 → .790	62.7 → 72.1
Framingham	64.1%	87.3%	90.0%	.678 → .706	51.3 → 51.2

TABLE X
AUROC UNDER TEST-TIME MISSING FEATURES. NATIVE = NATIVE MISSING-VALUE HANDLING; MED. = MEDIAN IMPUTATION; UNREG. = UNREGULARIZED TRANSFORMER.

Missing	SchemaFlexTabNet-QC	Unreg.	Native LGBM	Native XGB	Med. LGBM
0%	0.8520	0.8377	0.7297	0.7265	0.7274
10%	0.8300	0.8018	0.7035	0.7051	0.7090
20%	0.8036	0.7767	0.6854	0.6940	0.6944
30%	0.7510	0.7294	0.6718	0.6822	0.6779
40%	0.7159	0.6955	0.6412	0.6575	0.6539
50%	0.7123	0.6985	0.6131	0.6341	0.6337

clearer route to missing-feature robustness against native-NaN trees, and an abstention mechanism, but not a uniform win.

What it does not support. We do not show that duplicates explain the 90.1% in [1]; our evidence points the other way, and the origin of that figure remains undetermined. We do not show superiority over the strongest baseline on any cohort. Multi-seed variance and confidence intervals for the LOCO, ablation and conformal tables are outstanding.

A. Outcome-construct divergence

The pooled label mixes three constructs: *prevalent CVD* at a screening examination (Sulianova), *angiographic coronary disease* with stenosis verified at catheterization (UCI, Z-Alizadeh Sani), and *incident coronary events* over follow-up (Framingham) [5]. They share a binary label but not a clinical meaning, and a model cannot be expected to transfer equally well across them. This likely contributes to the weak and flat Framingham results, where AUROC stays near 0.68 for every model and conformal abstention does not improve accuracy. A cleaner design would use construct-specific heads or loss terms, or restrict cross-cohort claims to cohorts sharing an outcome definition.

Limitations.

1) *Outcome heterogeneity*, as discussed above.
2) *Pool composition.* Sulianova makes up 92.6% of training records, so learning of cardiology-specific variables relies on a few hundred patients. Inverse-frequency cohort weighting is a natural mitigation; its effect is not quantified here.
3) *Conformal validity.* Calibration data come from training cohorts and test data from a different cohort. Observed set coverage of 84.2–88.7% against a nominal 90% shows that guarantees do not transfer, so abstention here is empirical triage rather than a certified guarantee. Calibrating on labeled target data restores coverage but is unavailable zero-shot; covariate-shift weighting [17], [18] is a natural extension.
4) *Small evaluation cohorts* (≈300 patients) yield wide intervals, and most analyses are single runs.
5) *Baseline provenance.* HybridTabNet-QC is re-implemented and may differ from the authors' code. The tree baselines in the missingness test score lower at 0% missingness than the pooled cross-validated LightGBM, so their tuning should be checked before the absolute gaps are cited.
6) *Soft monotonicity* is measured, not guaranteed, and the 0.26% figure comes from synthetic perturbations of two features.
7) *Reproduction scope.* Our 70/15/15 reproduction uses our own pipeline and features; it shows that standard supervised learning on these features gives about 73%, not what the original authors' pipeline did.
8) *No external or prospective clinical validation.* The model is a research prototype, not a diagnostic tool.

IX. RECOMMENDED PRACTICE FOR CARDIOVASCULAR BENCHMARKS

Our audit suggests a short checklist that would have prevented the problems documented above and costs little to adopt.

1) *Deduplicate before splitting*, using exact and near-duplicate signature matching on clinical features, and report the number removed per source.
2) *Split at the patient level* and, where several files derive from one registry, check for overlap across files before any cross-cohort evaluation.
3) *Report the pooled composition*: the share of records contributed by each source, and the outcome definition used for each.
4) *Benchmark against the strongest tree baseline* with native missing-value handling and with the same engineered features as the proposed model.
5) *Evaluate on a held-out cohort*, not only on a random split of the pool, and give intervals and seed variance.
6) *Report conformal set coverage and acceptance rate separately*, and check coverage on a different cohort before claiming safety.
7) *Reconcile every reported number*: text, tables and figures (e.g. confusion matrices) should imply identical metrics.
8) *Publish code, splits and an audit log*, following TRIPOD+AI [22].

We emphasize that items 1–2 are enough, by themselves, to remove the most extreme inflation seen in public heart-disease files (Table III).

X. Conclusion

Headline accuracies on public cardiovascular benchmarks should be read with suspicion until duplicates, cross-cohort overlap and schema loss are audited. After cleaning, realistic accuracy on pooled routine data is about 73–76%, and a 70/15/15 reproduction of a published 90.1% setup gives about 73% whether or not the duplicates are kept. Cross-hospital transfer depends mostly on preserving unshared variables rather than on architectural novelty. SchemaFlexTabNet-QC offers a principled way to train across non-identical schemas, with a modest mean gain over a strong NaN-aware tree baseline that is not significant per cohort after Holm correction, graceful degradation under missing inputs (about 0.08–0.10 AUROC above native-NaN trees at 50% missingness), and an abstention mechanism whose coverage guarantee weakens under cohort shift. Its claims must be tightened by multi-seed evaluation and prospective validation. We release the audit log, schema registry and code to support independent verification.

Data and Code Availability

All datasets are public. Code, configuration, cleaned splits and result tables: https://github.com/RathanPaida/SchemaFlexTabNet-QC.

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